

On the Ocean Transports across the Nordic Seas Gateways

Does Fram Strait throttle Barents Sea Opening export, despite upstream supply via Greenland-Scotland Ridge?

Dong Jian · University of East Anglia · d.jian@uea.ac.uk · bve23zsu.github.io

Motivation

- The oceanic gateways to the Arctic are interconnected. Atlantic inflow across the Greenland–Scotland Ridge (GSR) carries heat toward the Barents Sea Opening (BSO) and Fram Strait (FS); Arctic-origin and dense waters return through FS and the GSR.
- FS and BSO volume transports are robustly anti-correlated across models (de Boer et al. 2018; Muilwijk et al. 2018) and observation-based inversions (Tsubouchi et al. 2018);
- Does FS control how much leaves BSO, despite GSR supplies inflow?

Data & methods

- A high-resolution model in the Nordic Seas (explore the model via [nordicseas3d.github.io](https://github.com/bve23zsu/nordicseas3d))
- control-volume budget across three gateways.
- composite analysis
- A hierarchy of observations:
 - Mooring observation
 - ERA5

Key Results

- Volume budget closure across GSR, BSO, and FS
- $1.1 \text{ Sv (GSR)} + 2.1 \text{ Sv (FS)} = 3.2 \text{ Sv (BSO)}$
- $r(\text{BSO, FS}) = -0.76$
- BSO and FS show lead-lag correlation peak at 0-1 month

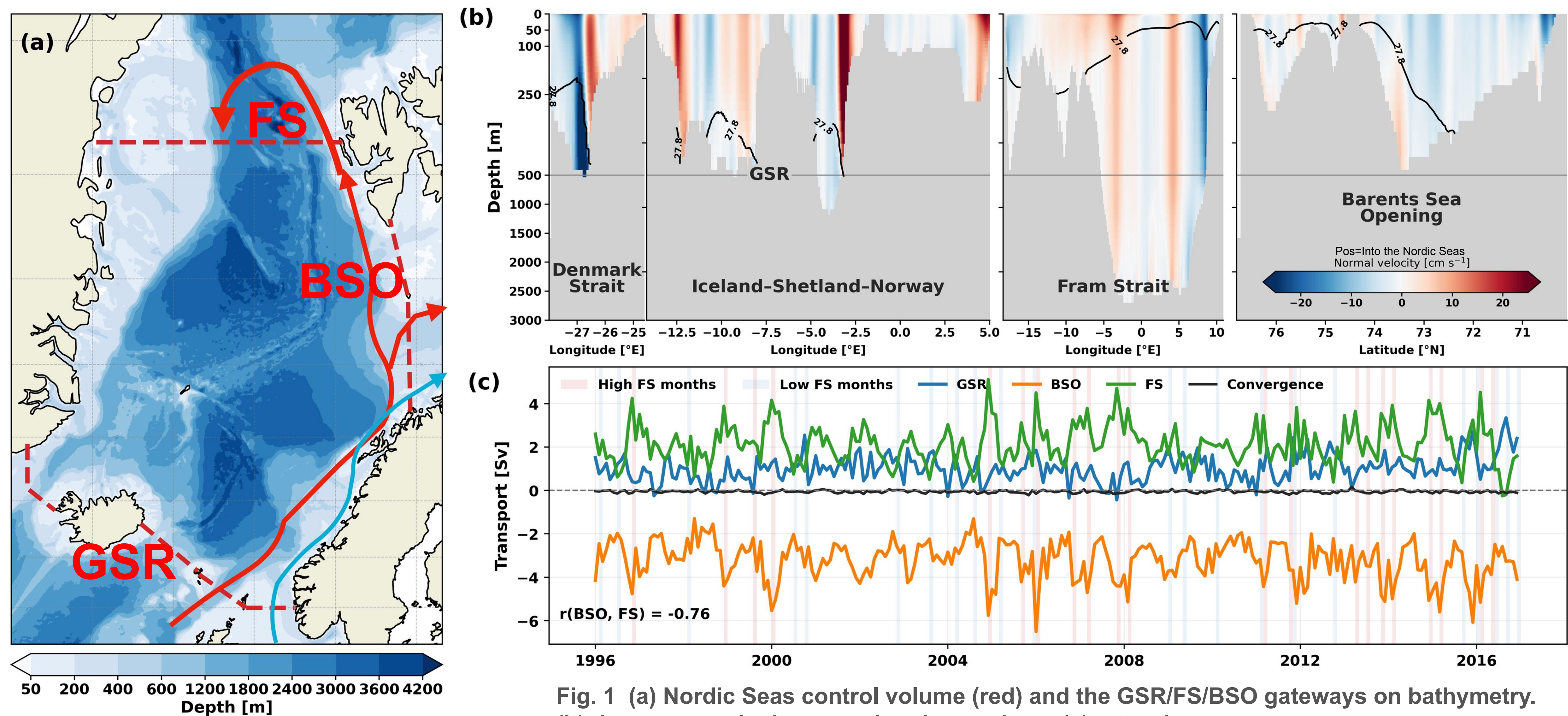


Fig. 1 (a) Nordic Seas control volume (red) and the GSR/FS/BSO gateways on bathymetry. (b) time-mean velocity normal to the sections. (c) net volume transport across gateways.

- High-low FS volume transport composite show possible associations

- Basin-wide cyclonic wind stress composite around Svalbard
- Composite SSH lower in the basin but is elevated in southern BSO
- Coherent gyre-scale anomaly in streamfunction spanning the basin
- Column-uniform changes in velocity; 27.8 isopycnal barely changes

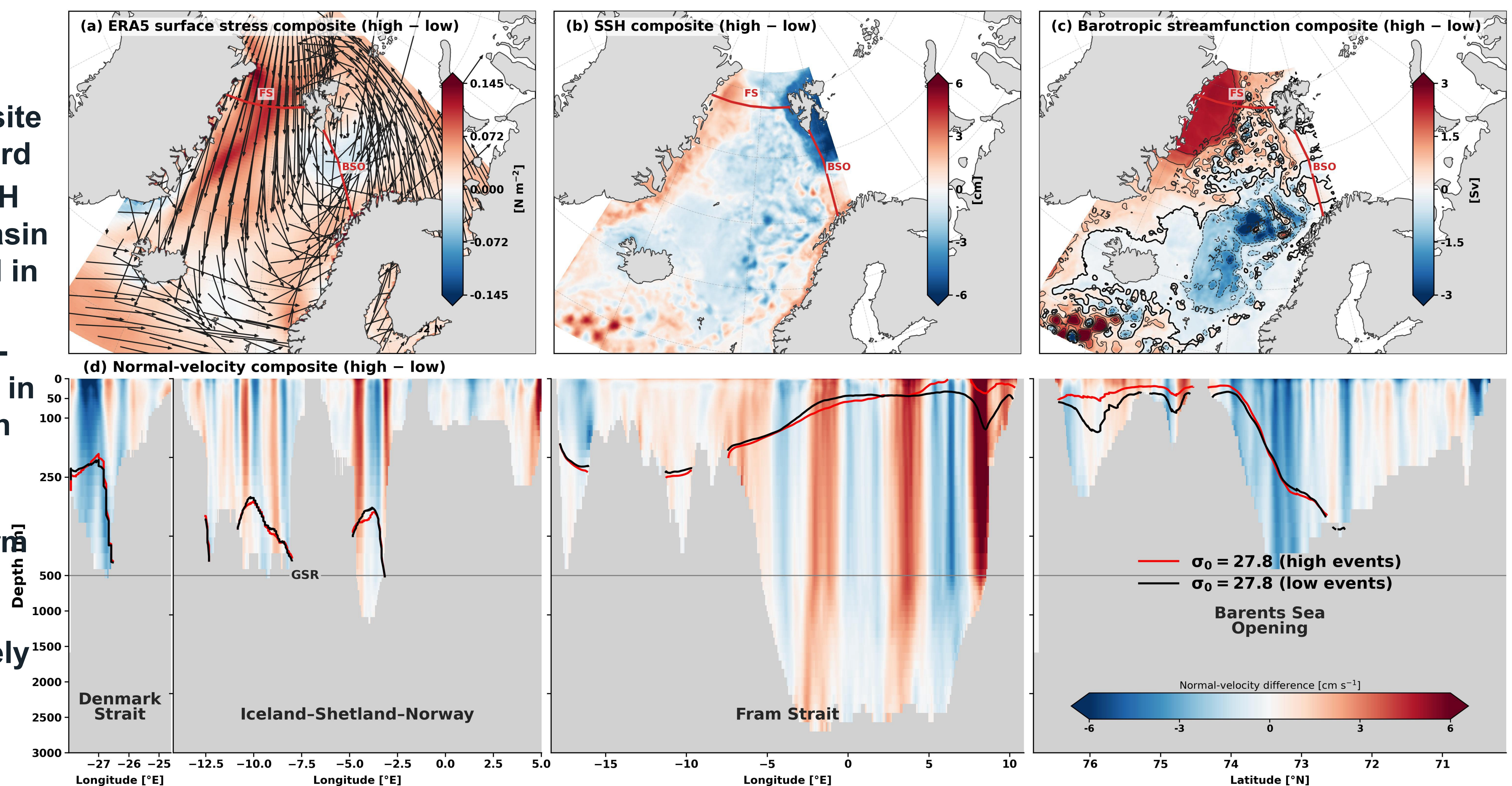


Fig. 2 Composite for (a) surface wind stress, (b) sea surface height, (c) barotropic streamfunction, and (d) velocity normal to the section.

Hypothesis and plans

- 1) BSO volume transport towards Barents Sea has been explained by upstream Atlantic inflow, we hypothesize that a coherent basin-wide barotropic circulation adjustment and associated wind pattern control the BSO transport, not merely upstream supply.
- 2) BSO export variability is compensated predominantly through Fram Strait; A synchronous coupling accompanies a basin-scale barotropic circulation adjustment, suggesting FS net volume import can possibly modulate BSO transport? Or the weather system above controls FS and BSO simultaneously?

Next, I will

- 1) Decompose transport into geostrophic and Ekman part
- 2) Examine moored observations at FS and BSO
- 3) Study synoptic cases